



TiDAS Capstone: Retrofitting Intelligence

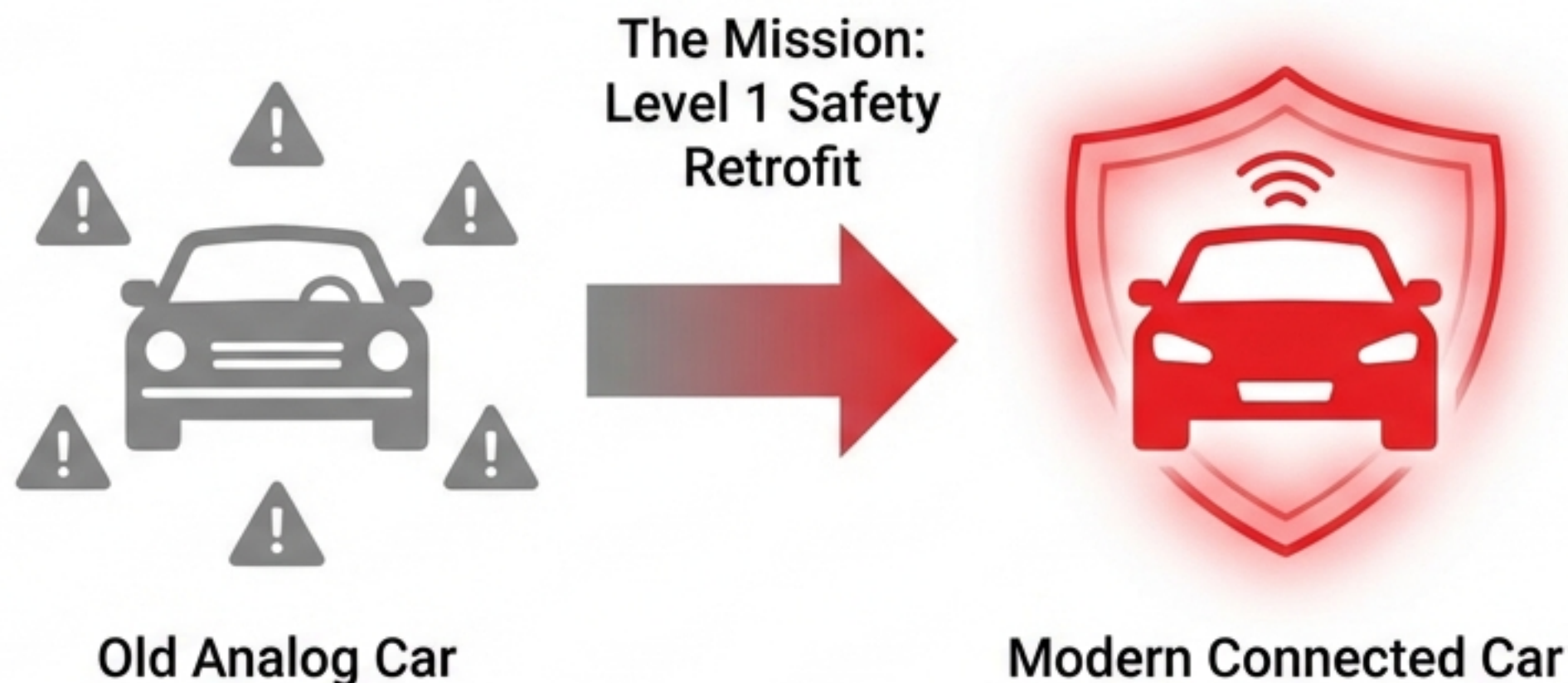
A Distributed Architecture for Aftermarket Safety

Team Name | January 22, 2026

The Safety Gap: 1.4 Billion Vehicles Left Behind

1.4 Billion

Older vehicles lack modern safety warnings. Current retrofits are unconnected and lack sensor fusion.



Engineering Constraints & Requirements



360° Coverage: Blind spots + Front collision.



Modularity: Adapts to variable sensor positions.



Privacy by Design: Local processing only (PIPEDA).

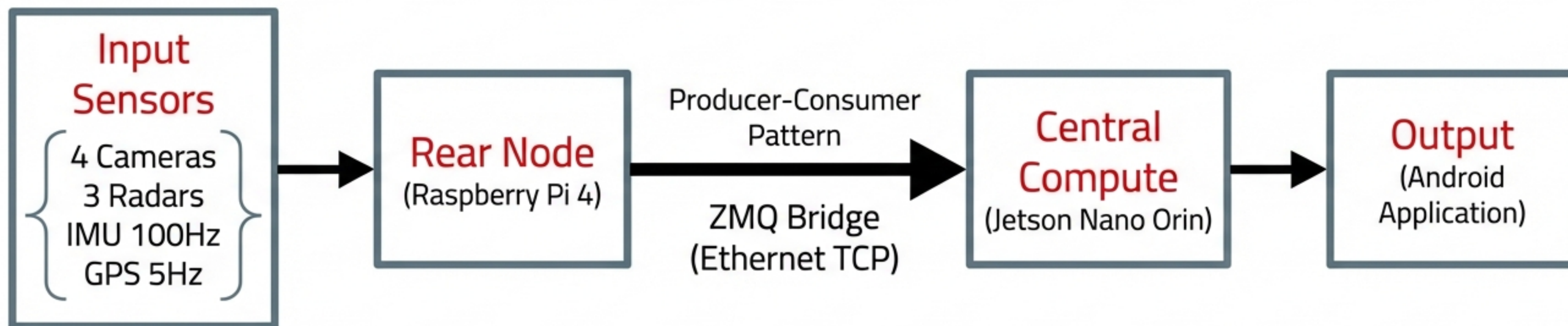


Reliability: Independent failure modes.

< 200ms

Latency Limit
(Human Reaction Buffer)

The Distributed Architecture



Design Trade-offs: Why Build Custom?

Rejected Approaches

Centralized USB: Bandwidth limits & signal degradation > 3m.

ROS (Robot OS): High overhead & latency.

LiDAR: Too slow (5.5Hz scan rate vs 60km/h travel).

Selected Approach

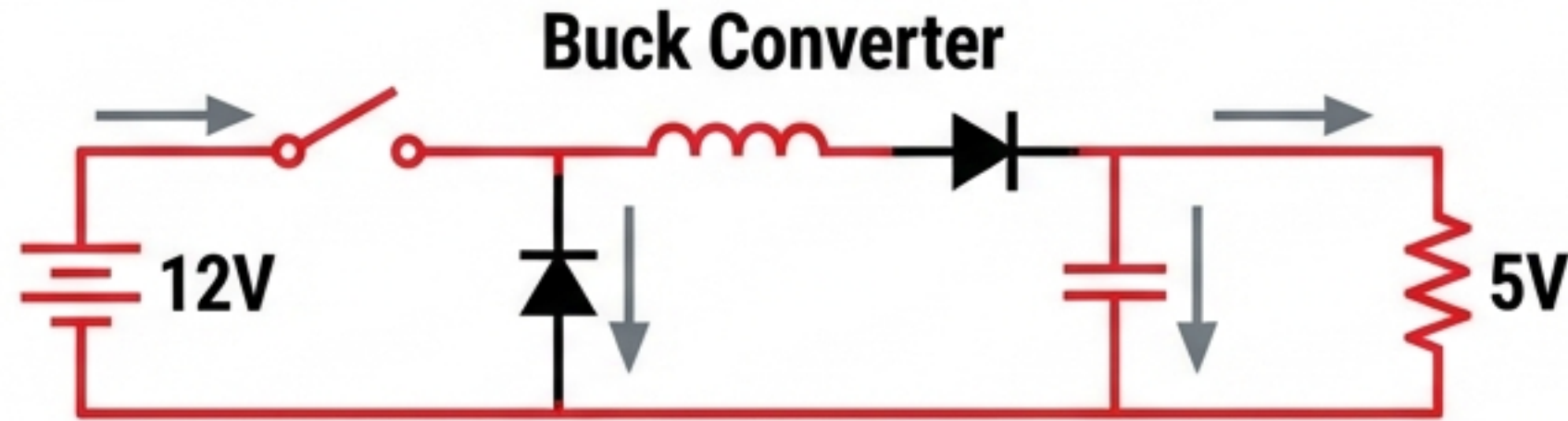
Distributed Custom C++: Low latency optimization.

ZMQ Bridge: Automatic framing & reconnection.

MJPEG Compression: Controlled edge bandwidth.

Physical Engineering: Power & Network

The Power Challenge



Problem: 5-Amp Spike. **Solution:** Custom Regulation.

The Network Integrity



USB (Degrades > 3m)

Static IP Configuration
& Chrony (NTP) Sync.



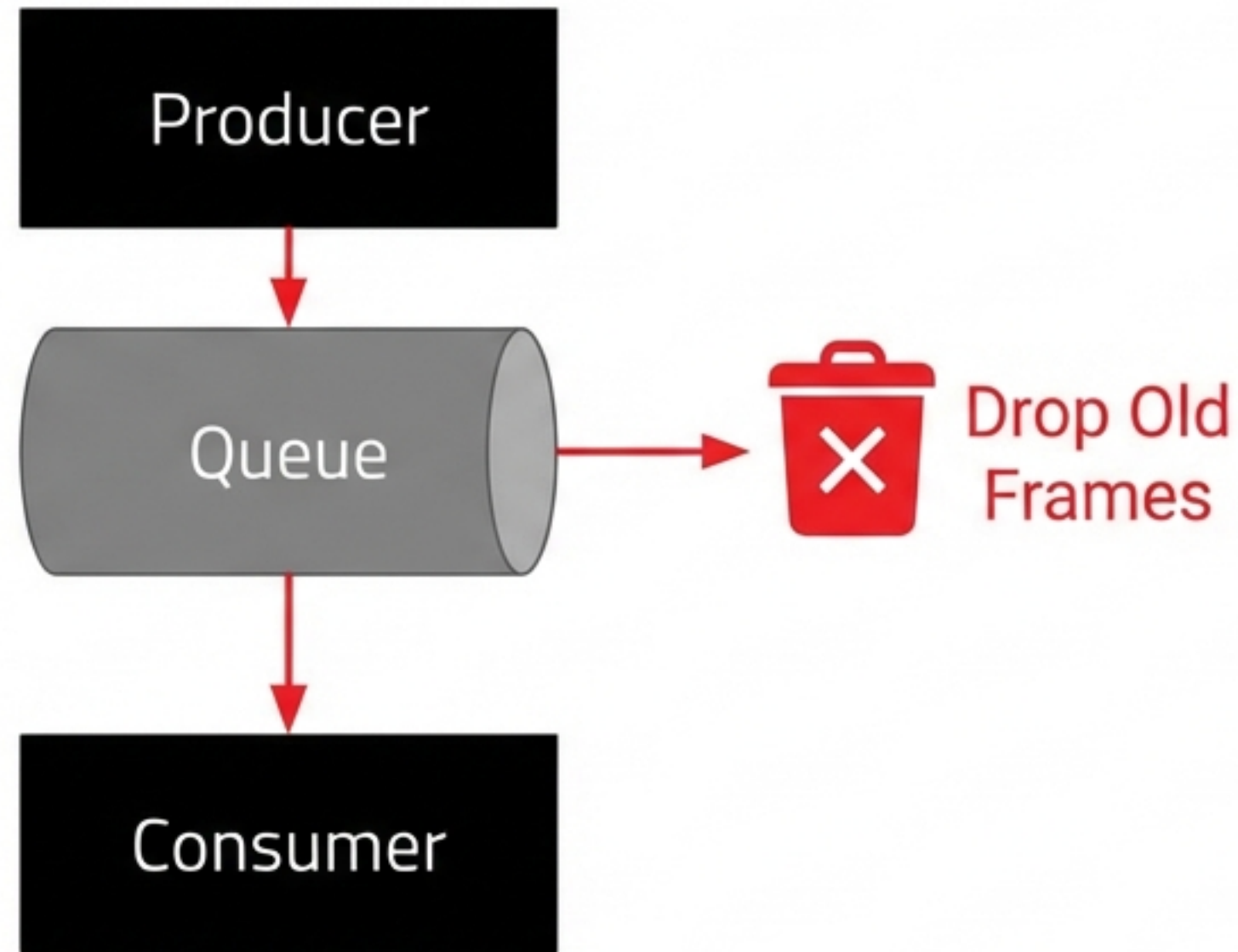
Static IP Configuration & Chrony (NTP) Sync.

Ethernet (TCP/ZMQ)

Implementation: The Perception Engine

Turning Raw Signals into Safety Warnings

Ingestion Strategy: Freshness over Completeness



Mechanism: Lock-free SPSC Queues.

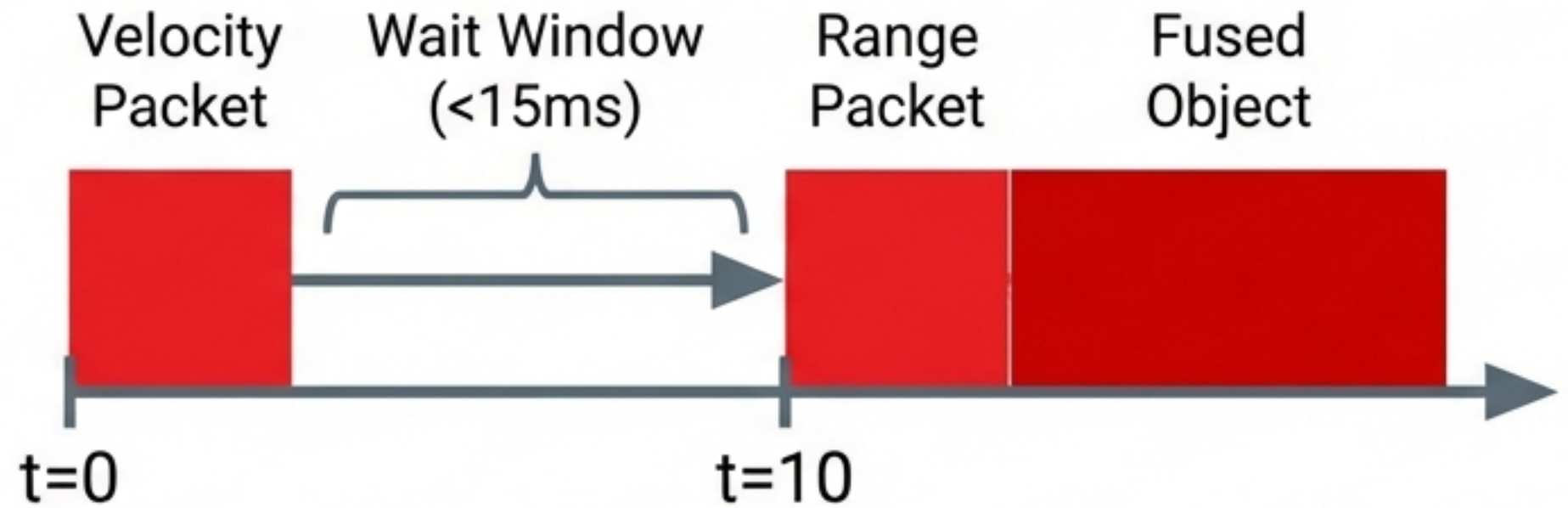
Logic: If AI lags, drop data. Processing 500ms old frames is unsafe.

Radar Physics & The “Split Packet” Problem

**Hardware: OPS243 FMCW
Doppler Radar**

- No Azimuth (Range/Velocity only).
- Sends Range and Velocity in separate packets.

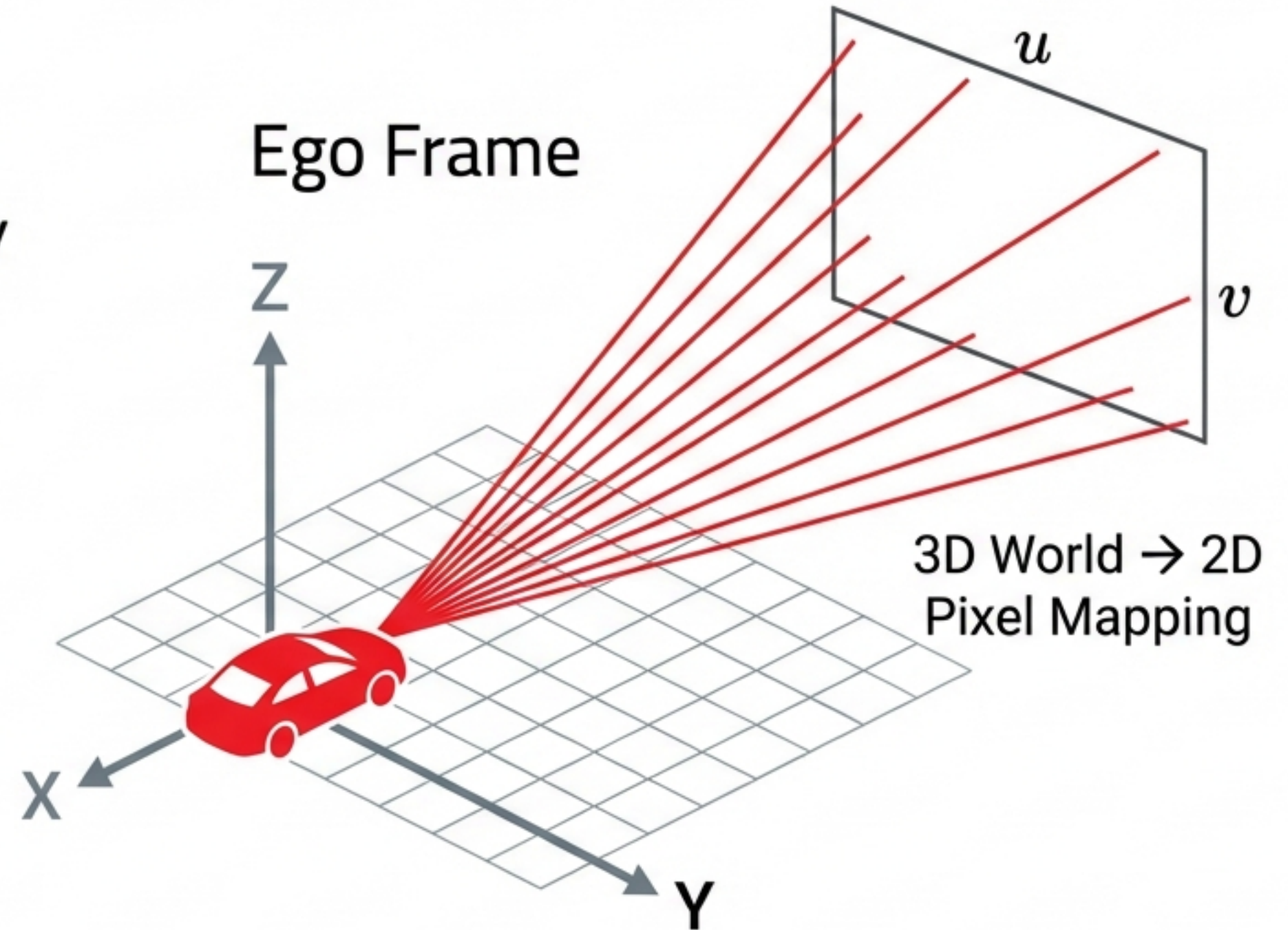
Algorithm: “Hold & Fuse”



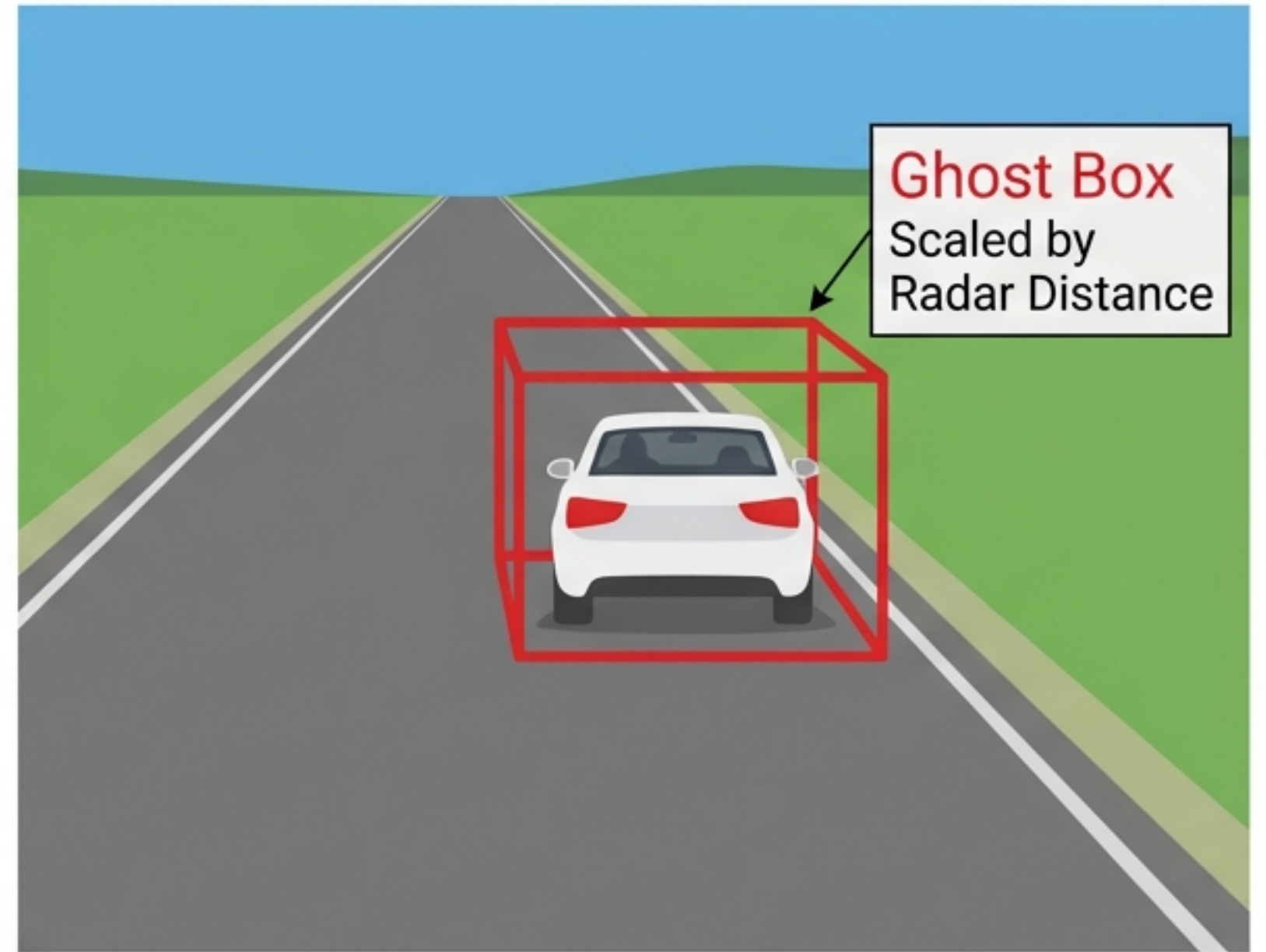
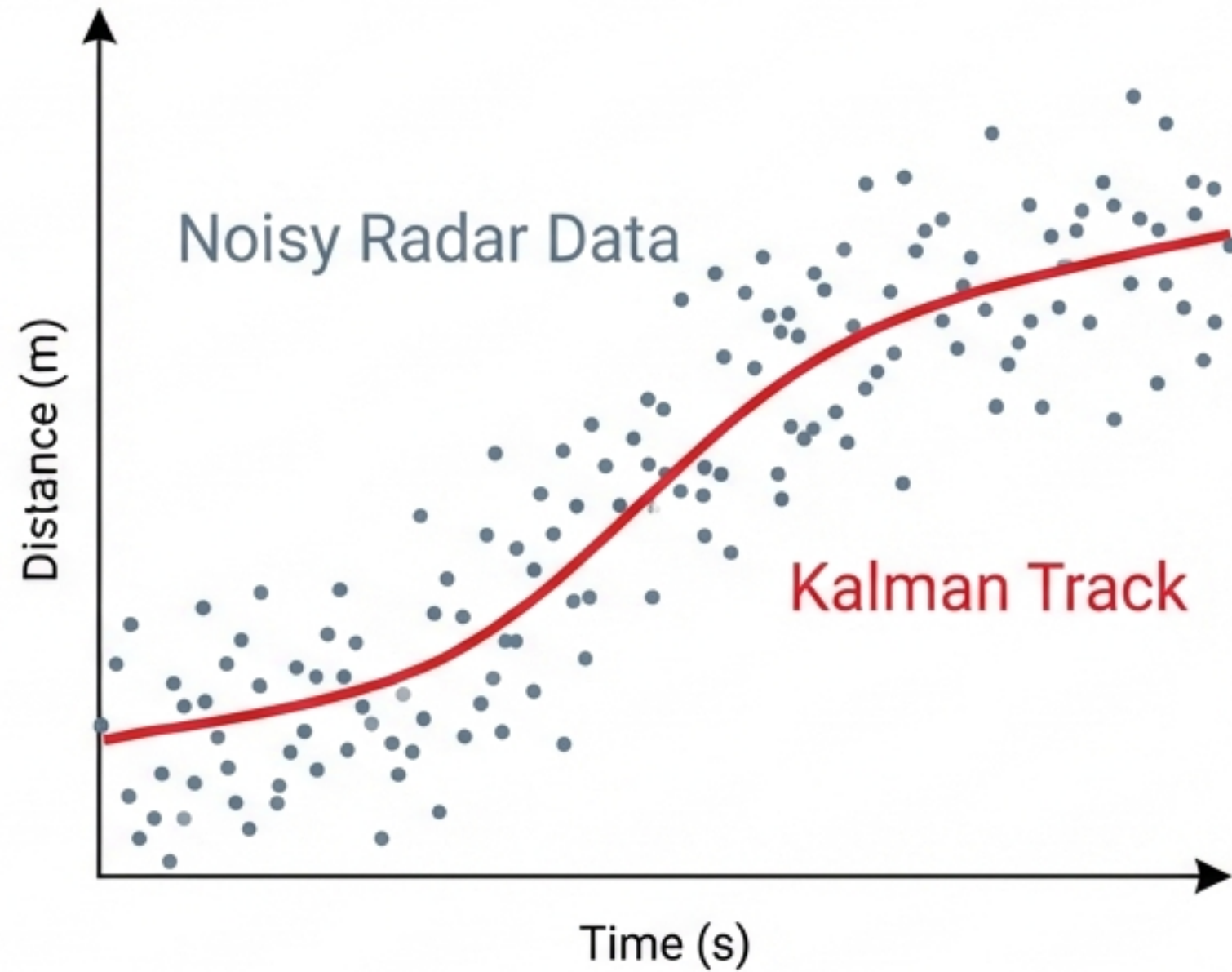
Kalman Filter (Constant Velocity) smooths noise.

Computer Vision & Spatial Transformation

- Model: YOLOv8 Nano (ONNX Optimized)
- Math: Homography Matrix / Inverse Pinhole Model



Sensor Fusion: The "Ghost Box"



FCW Trigger = **Negative Velocity** (Closing) + **TTC** < 2.5s

Connectivity: Optimizing BLE Bandwidth

Fragmentation Strategy

- Payload > MTU (240 bytes).
- Manual Fragmentation Header: [tick_id | seq_no | seq_max].
- Reassembly on Android Client.

CBOR Schema

Binary serialization for speed

Reliability: Watchdog Timer restarts pipeline if 'Tick' hangs > 1s.

User Experience: The 'Dark Cockpit'

Philosophy: Minimize Distraction.



Architecture: MVVM Pattern
(Decoupled UI/Data).



Parsing threads separated from UI
threads to prevent ANR.

Verification: The Vertical Slice

Stage	Latency
Sensor Capture	~30ms
Compute/Inference	~60ms
Transmission (BLE)	~40ms
Display Rendering	~20ms
Total System Latency	< 180ms (Passes 200ms Req)

Verified via high-speed camera timestamp analysis.

Conclusion & Roadmap

Summary: Engineered a modular, distributed platform meeting the 200ms safety threshold.

Future Roadmap (Jan - April):

1. Full Camera Calibration.
2. Complex Fusion (Lane Departure).
3. Robustness Testing.

Q&A